TEMASEK JUNIOR COLLEGE
Preliminary Examinations
Higher 1

CANDIDATE NAME

CIVICS GROUP: _________/10

INDEX NUMBER:

BIOLOGY 8875/02

Paper 2

12th September 2011

2 hours

Additional Materials: Paper

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** the questions.

Section B
Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of eah question or part question.

For Examiner's Use	
1	
2	
3	
4	
5 or 6	
Total	

Section A

Answer **all** the questions in this section.

1 Plants native to warm climates sometimes suffer injuries when exposed to relatively low temperatures. For example, temperatures in the range of 10℃ to 15℃ can cause "chilling injuries" to some sub-tropical plants that normally grow between 20℃ and 25℃. These chilling injuries may affect gene expression and reduce the rates of photosynthesis and protein synthesis. Some plants are more resistant to chilling injuries than others. These injuries could be due to damaged membranes.

Table 1.1 shows the fatty acid composition of mitochondrial membranes in various plants.

		Fatty acid composition / %					
		Chill-resistant plants			**Chill-sensitive plants**		
Fatty acid	No. of double bonds	Cauliflower	Turnip	Pea	Bean	Sweet potato	Maize
Palmitic	0	21.3	19.0	12.8	24.0	24.9	28.3
Stearic	0	1.9	1.1	2.9	2.2	2.6	1.6
Oleic	1	7.0	12.2	3.1	3.8	0.6	4.6
Linoleic	2	16.4	20.6	61.9	43.6	50.8	54.6
Linolenic	3	49.4	44.9	13.2	24.3	10.6	6.8
Ratio of unsaturated to saturated fatty acids		3.2:1	3.9:1	3.8:1	2.8:1	1.7:1	2.1:1

Source: A C Terry et al, Plant physiology 2000, 124, pages 183–190

Table 1.1

(a) (i) With reference to Table 1.1, describe the relationship between fatty acid composition and chill sensitivity.

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(ii) Explain how the presence of double bonds affects membrane integrity at low temperatures.

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(iii) Suggest how the function of mitochondria might be affected when a sweet potato plant is subjected to low temperatures.

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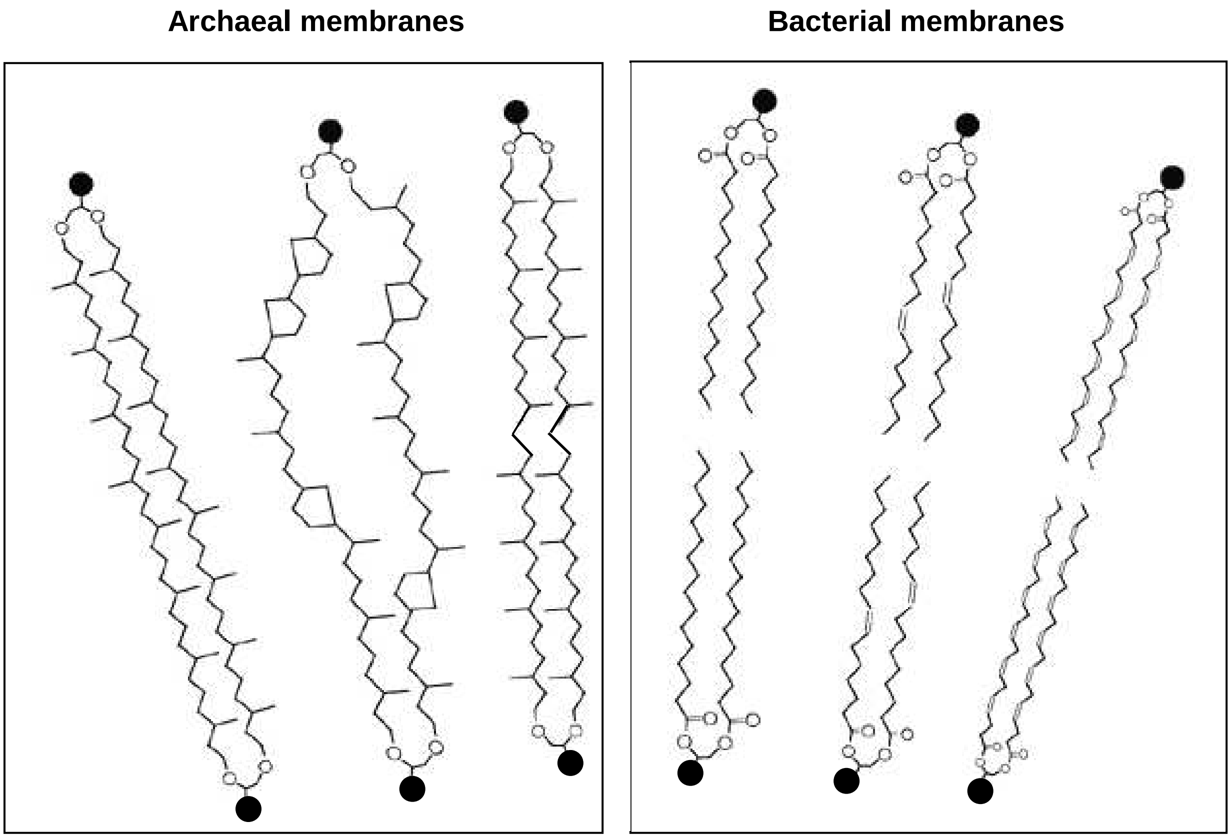
In the late 1970s, a new group of organisms, the Archaea was discovered. These single-celled organisms were found to inhabit extreme environments, such as salt lakes, hot springs and volcanic vents at temperatures of well over 100°C.

Archaea were initially thought to be structurally similar to bacteria. However, further work by scientists revealed major differences in the chemical compositions of their cell membranes. For example, the hydrocarbon tails of archaeal phospholipids consist of isoprenoid chains which are long chains with multiple side branches and rings. This is in contrast to the fatty acid chains which make up bacterial phospholipids.

For Examiner's Use

Fig. 1.1 shows the structure of the phospholipids in the membranes of archaea and bacteria.



Source: Nature Reviews Microbiology 5, 316-323 (April 2007)

Fig. 1.1

(b) (i) With reference to Fig. 1.1, other than the presence of side branches and rings, state two structural differences between the phospholipids of archaea and bacteria.

Archaea membranes	Bacterial membranes

[2]

For Examiner's Use

(ii) Suggest how the differences stated in (b)(i) enable archaea to thrive in environments with extreme conditions.

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[Total: 11]

2 C3 plants undergo photorespiration, a process by which RuBP has oxygen added to it by the enzyme RuBisCO, instead of carbon dioxide during normal photosynthesis. Photorespiration can occur when carbon dioxide levels are low and when temperatures are high, but unlike respiration, photorespiration produces no ATP.

Fig. 2.1 shows the rate of photosynthesis (measured by rate of CO_2 fixation, graph drawn in solid line) and the rate of photorespiration (measured by rate of O_2 fixation, graph drawn in dotted line) in wheat over a typical summer day.

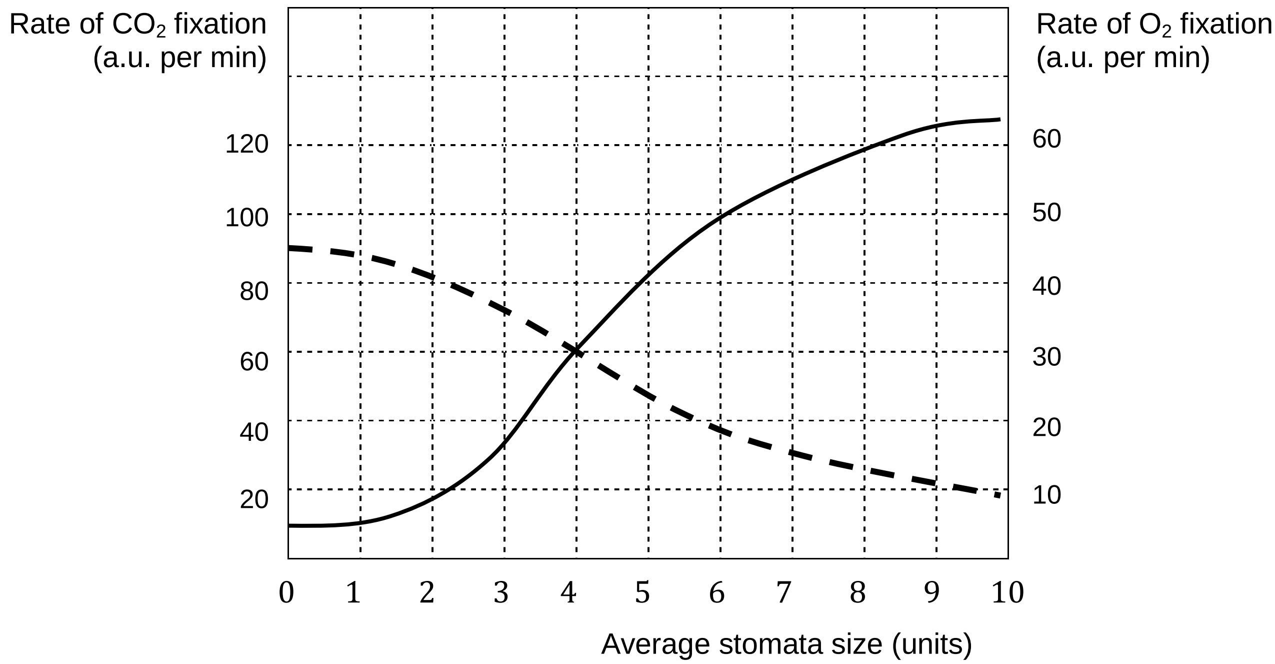

Fig. 2.1

(a) With reference to Fig. 2.1, explain how stomata size affects the overall rate of photosynthesis in wheat.

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For Examiner's Use

(b) State the type of inhibitor that oxygen is to the enzyme RuBisCO.

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Some plants have mechanisms to reduce photorespiration, such as the C4 plants. These plants undergo additional steps of capturing carbon dioxide using another enzyme (PEP carboxylase) to form oxaloacetate, before undergoing the Calvin cycle. PEP carboxylase has been found to have a lower K_m than RuBisCO for carbon dioxide. Examples of C4 plants include sugar cane and maize.

(c) Describe and explain the type of environment that C4 plants are usually found in.

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(d) Explain why Calvin cycle is still necessary in C4 plants.

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[Total: 9]

3 Back colour in the Kesametian Finch (Fig. 3.1) is determined by two alleles, blue and yellow. When pure breeding blue finches are mated with pure breeding yellow finches, the backs of the male progeny are always green, while the female progeny will have the same back colour as their fathers.

Fig. 3.1

(a) Using suitable symbols, construct both genetic diagrams to explain the results.

For Examiner's Use

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(b) Explain the above results.

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[Total: 8]

4 Antithrombin III (ATIII) is an important plasma protease inhibitor with a central role in the coagulation system. Deficiency of ATIII can cause patients to suffer recurrent venous thrombosis and pulmonary embolism.

Previous work indicated that the human liver is a rich source of ATIII and thus presumably of ATIII mRNA. The level of immuno-reactive ATIII mRNA in various rat liver samples was determined by radioassay. This was to choose the most appropriate source for the construction of cDNA clones for the study of these ATIII sequences.

Table 4.1 shows the relative abundance of ATIII mRNA in rat liver.

Source of tissue	**Relative abundance**
12-14 week old fetus	0.09
18-20 week old fetus	0.19
Adult	1.00

Table 4.1

(a) (i) With reference to Table 4.1, state and explain which source of tissue was used to construct cDNA clones.

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(ii) Briefly describe how the isolated mRNA can be used to synthesize the gene.

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(iii) Suggest why this method is preferred over isolating DNA from the liver for cloning.

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Fig. 4.1 shows a vector map of the plasmid vector pUC19. Restriction sites for restriction enzymes are indicated on the map.

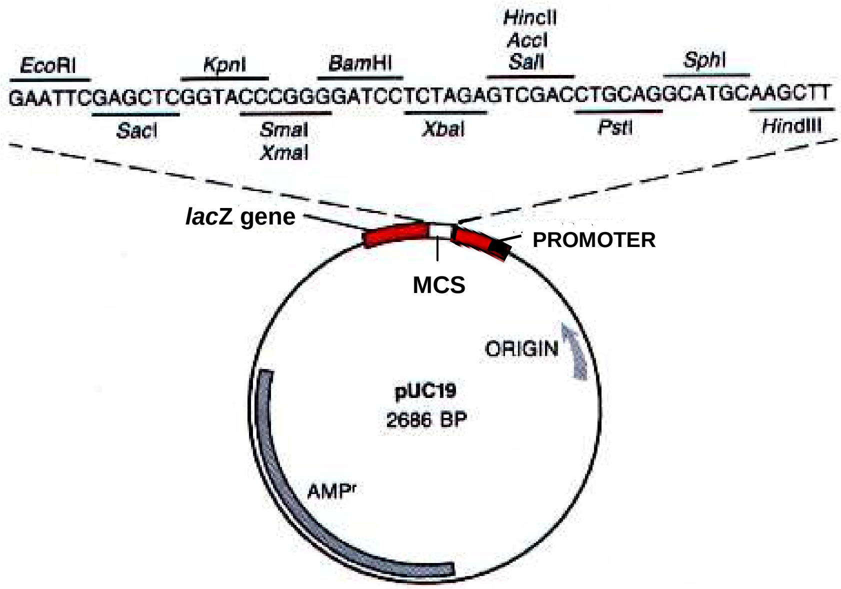

Fig. 4.1

(b) With reference to Fig. 4.1,

(i) explain the role of the MCS.

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(ii) explain why the "ORIGIN" and "AMP^r" are necessary in cloning the *ATIII* gene.

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One of the applications of the Human Genome Project (HGP) includes risk assessment of common diseases.

(c) Suggest how the HGP could potentially allow for the risk assessment of ATIII deficiency.

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[Total: 12]

Section B

Answer EITHER 5 OR 6.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a), (b)** etc., as indicated in the question.

Either

5 **(a)** Describe how different mature mRNA strands may be formed after transcription of a gene. [10]

(b) Describe the mode of action of enzymes with reference to its active site. [6]

(c) Outline how stem cells are used in cell-replacement therapy. [4]

[Total: 20]

Or

6 **(a)** Outline the functions of the membrane-bound organelles involved in protein synthesis and secretion. [5]

(b) State the functions of mitosis and describe the behaviour of chromosomes during the process. [8]

(c) Explain, with a named example, how environmental factors act as forces of natural selection. [7]

[Total: 20]